

MySQL

Exercise 1

1. Write a program that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, $\text{Perimeter} = 2*(L+W)$ and $\text{Area} = L*W$.

```
create procedure Rec1()
```

```
-> Begin
```

```
-> declare L int(4) default 10;
```

```
-> declare W int(4) default 5;
```

```
-> declare perimeter int(4);
```

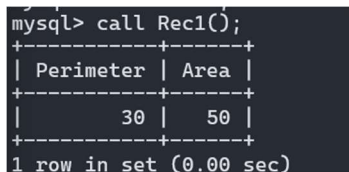
```
-> declare area int(4);
```

```
-> set perimeter=2*(L+W);
```

```
-> set area = L*W;
```

```
-> select Perimeter,Area ;
```

```
-> end//
```



```
mysql> call Rec1();
+-----+-----+
| Perimeter | Area |
+-----+-----+
|          30 |    50 |
+-----+-----+
1 row in set (0.00 sec)
```

2. Write a program that declares an integer variable called num, assigns a value to it, and computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
delimiter //
```

```
mysql> create procedure SquareCube(In x int(4))
```

```
-> begin
```

```
-> declare x2 int;
```

```
-> declare x3 int;
```

```

-> set x2=x*x*x;
-> set x2=x*x*x;
-> set x3=x*x*x*x;
-> set x2=x*x;
-> insert into tempp values(x,x2,x3);
-> end//

```

mysql> call SquareCube(4)//

```

mysql> select * from tempp
-> //
+-----+-----+-----+
| main_val | sq_val | cube_val |
+-----+-----+-----+
|      4 |     16 |      64 |
+-----+-----+-----+
1 row in set (0.01 sec)

```

mysql> call SquareCube(5)//

Query OK, 1 row affected (0.01 sec)

mysql> call SquareCube(6)//

Query OK, 1 row affected (0.01 sec)

mysql> call SquareCube(1)//

Query OK, 1 row affected (0.01 sec)

mysql> call SquareCube(2)//

Query OK, 1 row affected (0.00 sec)

```

mysql> select * from tempp//
+-----+-----+-----+
| main_val | sq_val | cube_val |
+-----+-----+-----+
|      4 |     16 |      64 |
|      5 |     25 |     125 |
|      6 |     36 |     216 |
|      1 |      1 |       1 |
|      2 |      4 |       8 |
+-----+-----+-----+

```

3. Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are:- $C = (F-32) \times 5/9$

$F = 9/5 \times C + 32$

```
mysql> create procedure ConvertTemp(In unit char(1),In temp decimal(10,2))
-> begin
->   declare toFar decimal(10,2);
->   declare toCel decimal(10,2);
->   if unit='F' then
->     set toCel=(temp-32)*5/9;
->     select concat(temp,'->') as Fahrenheit,toCel as Celsius;
->   else
->     set toFar=(temp*9/5)+32;
->     select concat(temp,'->') as Celsius,toFar as Fahrenheit;
->   end if;
-> end;
```

```
mysql> call ConvertTemp('F',100)//
```

Fahrenheit	Celsius
100.00->	37.78

1 row in set (0.04 sec)

```
mysql> call ConvertTemp('C',100)//
```

Celsius	Fahrenheit
100.00->	212.00

1 row in set (0.00 sec)

4. Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

5. Write a program that enables a user to input an integer. The program should then state whether the integer is evenly divisible by 5.

```
mysql> create procedure DivFive(In val int)
-> begin
->   declare rem int;
->   set rem = val%5;
->   if rem = 0 then
```

```
->      select concat(val,' is evenly divisible by 5') as result;
->  else
->      select concat(val,' is not evenly divisible by 5') as result ;
->  end if;
-> end; //
```

```
mysql> call DivFive(25)//
+-----+
| result |
+-----+
| 25 is evenly divisible by 5 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> call DivFive(23)//
+-----+
| result |
+-----+
| 23 is not evenly divisible by 5 |
+-----+
1 row in set (0.00 sec)
```

6. Your block should read in two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

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